=== UNIT 2: DIFFERENTIAL CALCULUS, LEIBNIZ THEOREM & CURVATURE ===

>> 2.1 Successive Differentiation and Leibniz’s Theorem

Differential calculus explores instantaneous rates of change, curvature, and series representations of engineering functions.
* Higher Order Derivatives: If y = f(x), then y_n = d^n y / dx^n. Standard nth derivatives:
- d^n/dx^n (e^{ax}) = a^n * e^{ax}.
- d^n/dx^n ((ax + b)^m) = m*(m-1)*...*(m-n+1) * a^n * (ax + b)^{m-n}.
- d^n/dx^n (sin(ax + b)) = a^n * sin(ax + b + n*pi/2).
- d^n/dx^n (cos(ax + b)) = a^n * cos(ax + b + n*pi/2).
- d^n/dx^n (ln(ax + b)) = (-1)^{n-1} * (n-1)! * a^n / (ax + b)^n.
* Leibniz’s Theorem on Product Differentiation: If u(x) and v(x) possess derivatives up to nth order, the nth derivative of their product y = u * v is given by the binomial-like expansion:
(u * v)_n = C(n,0)*u_n*v + C(n,1)*u_{n-1}*v_1 + C(n,2)*u_{n-2}*v_2 + ... + C(n,r)*u_{n-r}*v_r + ... + C(n,n)*u*v_n,
where C(n, r) = n! / [r! * (n-r)!].

>> 2.2 Mean Value Theorems and Taylor’s Expansions

* Rolle’s Theorem: If f(x) is continuous on [a, b], differentiable on (a, b), and f(a) = f(b), then there exists at least one c in (a, b) such that f’(c) = 0.
* Lagrange’s Mean Value Theorem (LMVT): If f(x) is continuous on [a, b] and differentiable on (a, b), there exists c in (a, b) such that f’(c) = [f(b) - f(a)] / (b - a).
* Cauchy’s Mean Value Theorem: For continuous f, g on [a, b] and differentiable on (a, b) with g’(x) != 0, there exists c in (a, b) such that [f(b) - f(a)] / [g(b) - g(a)] = f’(c) / g’(c).
* Taylor’s Series (Expansion about x = a):
f(x) = f(a) + (x - a)*f’(a) + [(x - a)^2 / 2!]*f’’(a) + ... + [(x - a)^n / n!]*f^{(n)}(a) + R_n(x).
* Maclaurin’s Series (Taylor series centered at a = 0):
f(x) = f(0) + x*f’(0) + (x^2 / 2!)*f’’(0) + (x^3 / 3!)*f’’’(0) + ... + (x^n / n!)*f^{(n)}(0) + ...

>> 2.3 Indeterminate Forms and L’Hopital’s Rule

* Standard 0/0 and inf/inf forms: lim_{x->a} [f(x)/g(x)] = lim_{x->a} [f’(x)/g’(x)], provided the limit on the right exists.
* Other Indeterminate Forms: 0 * inf (convert to (0)/(1/inf) = 0/0), inf - inf

(combine over common denominator), 0^0, 1^inf, inf^0 (evaluated by taking natural logarithms: y = f(x)^g(x) => ln(y) = g(x)*ln(f(x))).

>> 2.4 Curvature and Radius of Curvature (rho)

* Definition of Curvature (kappa): Rate of change of direction of tangent angle psi with respect to arc length s: kappa = d(psi) / ds.
* Radius of Curvature (rho): Reciprocal of curvature: rho = 1 / kappa = ds / d(psi).
* Cartesian Formula: rho = [1 + (y_1)^2]^{3/2} / | y_2 |, where y_1 = dy/dx and y_2 = d^2y/dx^2.
* Parametric Formula (x = x(t), y = y(t)): rho = [(x')^2 + (y')^2]^{3/2} / | x'*y'' - y'*x'' |.
* Polar Formula (r = f(theta)): rho = [r^2 + (r_1)^2]^{3/2} / | r^2 + 2*(r_1)^2 - r*r_2 |, where r_1 = dr/d(theta) and r_2 = d^2r/d(theta)^2.

[SOLVED PROBLEM 1: Application of Leibniz Theorem to Differential Equation]

Problem: If y = sin(m * arcsin(x)), prove that (1 - x^2)*y_{n+2} - (2n + 1)*x*y_{n+1} + (m^2 - n^2)*y_n = 0.
Solution Step 1: Differentiate once with respect to x:
y_1 = cos(m * arcsin(x)) * [m / sqrt(1 - x^2)].
Multiply by sqrt(1 - x^2): sqrt(1 - x^2) * y_1 = m * cos(m * arcsin(x)).
Solution Step 2: Square both sides to eliminate radicals:
(1 - x^2) * (y_1)^2 = m^2 * cos^2(m * arcsin(x)) = m^2 * [1 - sin^2(m * arcsin(x))].
Since y = sin(m * arcsin(x)), this simplifies to: (1 - x^2) * (y_1)^2 = m^2 * (1 - y^2).
Solution Step 3: Differentiate again with respect to x:
(1 - x^2) * [2*y_1*y_2] + (-2x) * (y_1)^2 = m^2 * [-2*y*y_1].
Divide through by common factor 2*y_1 (since y_1 != 0):
(1 - x^2)*y_2 - x*y_1 + m^2*y = 0.
Solution Step 4: Apply Leibniz's Theorem to differentiate n times:
Term 1: D^n [(1 - x^2) * y_2] = C(n,0)*(1 - x^2)*y_{n+2} + C(n,1)*(-2x)*y_{n+1} + C(n,2)*(-2)*y_n
= (1 - x^2)*y_{n+2} - 2n*x*y_{n+1} - n*(n - 1)*y_n.
Term 2: D^n [-x * y_1] = - [C(n,0)*x*y_{n+1} + C(n,1)*(1)*y_n] = -x*y_{n+1} - n*y_n.
Term 3: D^n [m^2 * y] = m^2 * y_n.
Solution Step 5: Collect terms:
(1 - x^2)*y_{n+2} + [-2n*x - x]*y_{n+1} + [-n^2 + n - n + m^2]*y_n = 0
=> (1 - x^2)*y_{n+2} - (2n + 1)*x*y_{n+1} + (m^2 - n^2)*y_n = 0. (Hence Proved).

[SOLVED PROBLEM 2: Taylor Series Expansion of Function]

Problem: Expand f(x) = e^x in powers of (x - 2) up to the fourth-degree term and approximate e^{2.1}.
Solution Step 1: Evaluate derivatives at center a = 2:
f(x) = e^x => f(2) = e^2
f'(x) = e^x => f'(2) = e^2
f''(x) = e^x => f''(2) = e^2
f'''(x) = e^x => f'''(2) = e^2
f^{(4)}(x) = e^x => f^{(4)}(2) = e^2.
Solution Step 2: Substitute into Taylor's formula:
e^x = e^2 + e^2*(x - 2) + (e^2 / 2!)*(x - 2)^2 + (e^2 / 6)*(x - 2)^3 + (e^2 / 24)*(x - 2)^4 + ...
e^x = e^2 * [1 + (x - 2) + 0.5*(x - 2)^2 + 0.1667*(x - 2)^3 + 0.0417*(x - 2)^4].
Solution Step 3: Evaluate at x = 2.1 (so x - 2 = 0.1):
Bracket = 1 + 0.1 + 0.5*(0.01) + 0.1667*(0.001) + 0.0417*(0.0001)
= 1 + 0.1 + 0.005 + 0.0001667 + 0.0000042 = 1.1051709.
Using e^2 = 7.389056:
e^{2.1} ~ 7.389056 * 1.1051709 = 8.16617.
Exact Value Check: e^{2.1} = 8.1661699 (Accurate to 6 decimal places).

[SOLVED PROBLEM 3: Indeterminate Form Evaluation (1^inf)]

Problem: Evaluate the limit L = lim_{x->0} [(tan(x) / x)]^{1 / x^2}.
Solution Step 1: Form identification: As x -> 0, tan(x)/x -> 1 and 1/x^2 -> inf. This is the 1^inf indeterminate form.
Solution Step 2: Take natural logarithm: ln(L) = lim_{x->0} [ln(tan(x) / x) / x^2] (0/0 form).
Solution Step 3: Apply L'Hopital's Rule:
Numerator derivative: d/dx [ln(tan(x)) - ln(x)] = (sec^2(x) / tan(x)) - (1 / x) = (1 / (sin(x)*cos(x))) - (1 / x)
= [2 / sin(2x)] - [1 / x] = [2x - sin(2x)] / [x * sin(2x)].
Denominator derivative: d/dx [x^2] = 2x.
ln(L) = lim_{x->0} [2x - sin(2x)] / [2x^2 * sin(2x)].
Solution Step 4: Expand sin(2x) using Maclaurin series sin(u) = u - u^3/6 + ...:
sin(2x) = 2x - (2x)^3 / 6 = 2x - (8x^3 / 6) = 2x - (4/3)*x^3.
Numerator = 2x - [2x - (4/3)*x^3] = (4/3)*x^3.
Denominator = 2x^2 * [2x] = 4x^3.
ln(L) = lim_{x->0} [(4/3)*x^3 / 4x^3] = (4/3) / 4 = 1/3.
Solution Step 5: Solve for L: L = e^{ln(L)} = e^{1/3}.
Final Answer: Limit = e^{1/3} ~ 1.3956.

[SOLVED PROBLEM 4: Radius of Curvature Calculation]

Problem: Find the radius of curvature rho at the vertex (a, 0) of the catenary curve y = a * cosh(x / a).
Solution Step 1: Calculate first derivative y_1:
y_1 = dy/dx = a * sinh(x / a) * (1 / a) = sinh(x / a).
At x = 0 (vertex, where y = a*cosh(0) = a): y_1(0) = sinh(0) = 0.
Solution Step 2: Calculate second derivative y_2:
y_2 = d^2y/dx^2 = cosh(x / a) * (1 / a) = (1 / a) * cosh(x / a).
At x = 0: y_2(0) = (1 / a) * cosh(0) = 1 / a.
Solution Step 3: Substitute into Cartesian curvature formula:
rho = [1 + (y_1)^2]^{3/2} / | y_2 |
rho = [1 + 0^2]^{3/2} / [1 / a] = 1 / (1 / a) = a.
General Point Formula: For any point (x, y) on the catenary:
rho = [1 + sinh^2(x / a)]^{3/2} / [(1 / a)*cosh(x / a)] = [cosh^2(x / a)]^{3/2} / [(1 / a)*cosh(x / a)]
= a * cosh^2(x / a) = y^2 / a.
Final Answer: At the vertex (0, a), rho = a. At any point (x, y), rho = y^2 / a.

>> 2.5 University Review & Examination Questions

* Q1: State and prove Lagrange’s Mean Value Theorem and give its geometric interpretation.
Answer: Statement: If f(x) is continuous on [a, b] and differentiable on (a, b), there exists at least one c in (a, b) such that f’(c) = [f(b) - f(a)] / (b - a). Proof: Define auxiliary function phi(x) = f(x) - A*x, choosing constant A such that phi(a) = phi(b). This yields A = [f(b) - f(a)] / (b - a). phi(x) satisfies all three conditions of Rolle’s Theorem on [a, b]. Hence, there exists c in (a, b) such that phi’(c) = 0. Since phi’(x) = f’(x) - A, phi’(c) = 0 implies f’(c) = A = [f(b) - f(a)] / (b - a).
Geometrically, this means there is at least one point on the curve where the tangent line is strictly parallel to the secant line joining endpoints (a, f(a)) and (b, f(b)).
* Q2: Derive the formula for the radius of curvature in polar coordinates r = f(theta).
Answer: In polar coordinates, x = r*cos(theta) and y = r*sin(theta). Differentiating with respect to theta yields dx/d(theta) = r’*cos - r*sin and dy/d(theta) = r’*sin + r*cos. The arc differential ds^2 = dx^2 + dy^2 = (r^2 + (r’)^2)*d(theta)^2, giving ds/d(theta) = sqrt(r^2 + (r’)^2). The angle psi made by the tangent with the polar axis satisfies tan(psi) = (dy/d(theta)) / (dx/d(theta)). Differentiating psi with respect to theta and using rho = ds / d(psi) = (ds/d(theta)) / (d(psi)/d(theta)) yields the standard polar formula: rho = [r^2 + (r_1)^2]^{3/2} / [r^2 + 2*(r_1)^2 - r*r_2], where r_1 = dr/d(theta) and r_2 = d^2r/d(theta)^2.

>> 2.6 Key Takeaways and Revision Points

* Leibniz’s theorem expands the nth derivative of a product analogously to the binomial theorem.

* Taylor and Maclaurin series approximate complex nonlinear engineering functions as power series.
* Indeterminate forms of type 1^inf, 0^0, inf^0 are solved by logarithmic transformation before applying L'Hopital's rule.
* Curvature measures rate of turning of the tangent vector; radius of curvature rho = [1 + (y')^2]^{3/2} / |y''|.